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EcoSpec

Beijing Institute of Technology Identifies Expert Scattering as a Fundamental Barrier to Speculative Decoding in MoE Models and Achieves Up to 1.62× Speedup Through Cost-Aware Draft Selection

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Speculative decoding accelerates autoregressive generation by drafting multiple tokens quickly and verifying them in a single forward pass through the large target model. For dense models this works cleanly: the verification pass has a fixed memory footprint. But for Mixture-of-Experts models — which power all current frontier open-source systems including DeepSeek-V3.1 and Qwen3 — researchers from Beijing Institute of Technology have identified a previously uncharacterized failure mode: *expert scattering*. Their remedy, EcoSpec, jointly optimizes acceptance likelihood and expert reuse, delivering up to 1.62× end-to-end speedup over standard speculative decoding at no quality cost.

AT A GLANCE

Problem: Confidence-driven draft selection in MoE models routes high-probability tokens to disjoint expert sets, increasing expert-weight memory traffic and eliminating the speedup from speculation.

Why it matters: All major frontier open-source models (DeepSeek-V3.1, Qwen3-235B) use MoE; speculative decoding must account for expert memory locality to be effective at scale.

Method: EcoSpec: lightweight expert predictor + dynamic expert buffer; composite draft scoring balancing acceptance likelihood and expert reuse fraction.

Result: Up to 1.62× end-to-end speedup on DeepSeek-V3.1 (671B) across reasoning, coding, QA, and dialogue benchmarks.

Evidence: Standard speculative decoding achieves negative speedup at draft length ≥ 8 on DeepSeek-V3.1; EcoSpec reaches 1.62× at the same draft length.

The Big Picture

Speculative decoding has become the standard technique for accelerating large language model inference: a small

draft model generates a sequence of candidate tokens at low cost, and the large target model verifies the entire sequence in a single forward pass, accepting tokens that agree with its own distribution and rejecting the rest. For dense models, each forward pass has a fixed compute and memory cost, and the speedup is determined solely by the acceptance rate of the draft tokens.

MoE models break this assumption. In an MoE model, each token is routed to a small subset of experts (typically 2 out of 64), and the memory footprint of a forward pass is proportional to the number of *unique* experts activated across all tokens in the batch. When draft tokens are selected to maximize acceptance likelihood alone, those tokens may route to many different expert sets — forcing the serving system to load additional expert weight matrices into GPU high-bandwidth memory for each verification step. This expert scattering can make speculative decoding slower than standard autoregressive generation.

Why Existing Approaches Fall Short

Prior speculative decoding methods for MoE models adopt the same draft selection criterion as for dense models: the draft token with highest likelihood under the draft model is selected at each position. This ignores the routing structure of the target model entirely.

Two prior attempts at MoE-aware speculative decoding — MoESD and Utility-Driven Speculative Decoding — optimize the expert activation pattern for the draft model but not for the target model, leaving the expert scattering problem at the target verification pass unresolved.

Core Mechanism

EcoSpec modifies draft selection by introducing two components. The **expert predictor** is a lightweight auxiliary network that, given the draft model’s hidden state for a candidate token, forecasts which of the target model’s 64 experts that token will activate. The **dynamic expert buffer** tracks which expert weights are currently cached in GPU high-bandwidth memory for the active verification batch. Draft selection then uses a composite score that balances the probability of token acceptance with the fraction of required experts already in the buffer, so the verification step can proceed with minimal additional expert loading.

Technical Formulation

Let c_i be a candidate token at draft position i and let $p_{\text{accept},i}$ be its acceptance probability under the target model. The expert predictor h_ψ maps the draft hidden

state to a binary vector over the expert set:

$$\hat{e}_i = h_\psi(d_i) \in \{0, 1\}^E$$

where d_i is the draft model’s hidden state at position i and E is the total number of experts. The expert reuse fraction is:

$$f_{\text{reuse},i} = \frac{|\hat{e}_i \cap \mathcal{B}|}{|\hat{e}_i|}$$

where $\mathcal{B}$ is the current expert buffer (the set of experts in GPU HBM). The composite draft score is:

$$s_i = p_{\text{accept},i} \cdot (1 + \lambda \cdot f_{\text{reuse},i})$$

with $\lambda = 0.4$ chosen by grid search. After each verification step, $\mathcal{B}$ is updated to include the experts activated by the accepted tokens.

Learning Procedure

1. **Expert predictor training:** h_ψ is trained offline on a 10K-sample calibration corpus to minimize prediction error for target expert activation patterns. Training takes approximately 30 minutes on a single GPU and adds <0.3% latency to each draft step.
2. **Draft model selection:** Any standard speculative decoding draft model can be used; EcoSpec modifies only the token selection step, not the draft model itself.
3. **Buffer initialization:** At the start of each generation, the buffer $\mathcal{B}$ is initialized with the experts activated by the prompt tokens’ verification pass.
4. **No target model modification:** EcoSpec requires no changes to target model weights, architecture, or serving infrastructure beyond the expert predictor and buffer tracking.

Experimental Findings

Model	Task	Std. SD	EcoSpec
DeepSeek-V3.1	Reasoning	1.00×	1.62×
DeepSeek-V3.1	Coding	1.00×	1.51×
Qwen3-235B-A22B	QA	1.00×	1.44×
GPT-OSS-120B	Dialogue	1.00×	1.38×

Standard speculative decoding achieves negative speedup (slower than autoregressive) on DeepSeek-V3.1 at draft lengths of 8 or more tokens, confirming expert scattering as a practical barrier. EcoSpec reduces the active expert footprint by 18–35% relative to confidence-only draft selection while maintaining or improving token acceptance rates.

Ablations and Interpretation

Expert predictor quality: Using a perfect oracle predictor improves speedup to $1.81\times$ on DeepSeek-V3.1, indi-

cating that the expert predictor is the current bottleneck and that improving its accuracy is the most promising direction for future gains.

Dynamic vs. static buffer: A static expert buffer (fixed set, not updated per step) achieves only $1.22\times$ speedup — less than half the gain of the dynamic variant — confirming that tracking live HBM state is essential for effective cost-aware draft selection.

Lambda sensitivity: Speedup on DeepSeek-V3.1 varies from $1.41\times$ ($\lambda = 0$, pure confidence) to $1.62\times$ ($\lambda = 0.4$) to $1.48\times$ ($\lambda = 1.0$, pure reuse). The optimal λ is consistent across models and tasks, suggesting that a moderate balance between confidence and reuse is universally beneficial.

Acceptance rate: EcoSpec’s composite scoring reduces the acceptance rate by at most 1.2 percentage points relative to confidence-only selection on any tested benchmark, confirming that the speedup comes from reduced expert loading rather than a higher acceptance rate — and that the quality of generated text is unchanged.

Reference

Jincheng Xie, Runheng Liu, Heyan Huang, Yawen Ling, Hanbin Dai, Yu Zheng, Wen Hu. **Less Experts, Faster Decoding: Cost-Aware Speculative Decoding for Mixture-of-Experts.** arXiv:2607.12696, July 2026. <https://arxiv.org/abs/2607.12696>